

Laboratoriya 12

Mavzu: DOM o'rqali elementlarni qoshish va o'zgartirish.

Ish Maqsadi:

Ushbu laboratoriya ishining asosiy maqsadi — JavaScript yordamida DOM manipulyatsiyasi asosiy usullarini amalda o'zlashtirish. Talabalar quyidagi ko'nikmalarni egallashlari lozim:

Elementlarni Yaratish: `document.createElement()` funksiyasidan foydalanib, HTML hujjatiga dasturiy yo'l bilan yangi teglar yaratish.

Elementlarni Qo'shish: Yaratilgan yoki mavjud elementlarni DOM daraxtida kerakli joyga (masalan, ro'yxatga yangi element qo'shish) `appendChild()` yoki `insertBefore()` kabi metodlar orqali joylashtirish.

Elementlarni O'zgartirish: Mavjud elementlarning matn kontentini (`.textContent` yoki `.innerHTML`) va ularning atributlarini (`.setAttribute()`) o'zgartirish.

Elementlarni O'chirish/Almashtirish: Keraksiz DOM tugunlarini sahifadan olib tashlash (`.remove()` yoki `.removeChild()`).

Kirish Qism:

Veb-sahifalar faqatgina statik HTML matnlaridan iborat emas. Zamonaviy interaktiv ilovalar sahifaning mazmunini va tuzilishini dinamik tarzda o'zgartira oladi. Aynan shu jarayonni boshqarish uchun JavaScript tiliga maxsus interfeys kerak bo'ladi. Bu interfeys DOM (Document Object Model) deb ataladi.

DOM — bu brauzerning HTML va XML hujjatlarini daraxtsimon tuzilma (Node Tree) sifatida ifodalash usuli. Bu daraxtdagi har bir HTML tegi, matn, hatto atribut ham alohida tugun (Node) sifatida ko'riladi. JavaScript DOMga murojaat qilish orqali HTML tuzilmasini o'qish, undan elementlarni olib tashlash, yangi elementlarni qo'shish yoki mavjud elementlarning uslublari (CSS) va kontentini o'zgartirish imkonini beradi.

Nazariy Qism:

A. DOM Tushunchasi

DOM birinchi navbatda, HTML hujjatini turli obyektlarning ierarxik to'plami sifatida ko'radi. Bu ierarxiyaning tepasida **document** obyekti turadi, u orqali barcha sahifa elementlariga kirish mumkin.

B. Element Yaratish va Konfiguratsiya qilish

Yangi element yaratish DOM manipulyatsiyasining birinchi qadamidir.

		const newDiv = document.createElement('div');
		newDiv.setAttribute('id', 'element-1');

C. Elementlarni DOMga Qo'shish

Yaratilgan element sahifada ko'rinishi uchun uni mavjud elementning farzandi yoki oldi/ortiga qo'shish kerak.

1. **parentNode.appendChild(childNode):** Eng ko'p ishlatiladigan metod. parentNode ichidagi **eng oxirgi farzand** sifatida childNode ni qo'shadi.
2. **parentNode.insertBefore(newChild, referenceChild):** referenceChild oldidan newChild ni qo'shadi.

D. Elementlarni O'zgartirish (Kontent va Atributlar)

Mavjud elementlarni topish va ularni yangilash uchun turli usullar mavjud.

	Element ichidagi butun HTML kontentini almashtirish (teglar bilan).	myElement.innerHTML = 'Yangi matn';
	Element ichidagi faqat matnni almashtirish (xavfsizroq usul).	Element.textContent = 'Oddiy matn.';
	Elementning inline CSS uslubini o'zgartirish.	

Elementni DOM daraxtidan olib tashlash quyidagicha amalga oshiriladi:

- **childNodes.remove():** Elementning o'zini o'chirishning eng yangi va sodda usuli (ES6+).
- **parentNode.removeChild(childNode):** Elementni uning ota-tuguni orqali o'chirish (eski brauzerlar uchun ham mos).

Amaliy qism:

Html kod:

```
<!DOCTYPE html>
<html lang="uz">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>12-Laboratoriya</title>
  <link rel="stylesheet" href="style.12lab.css">
</head>

<body id="bg">
  <div class="container">
    <h2>Foydalanuvchi Ma'lumotlari</h2>

    <div class="input-group">
      <label for="name">Ism:</label>
      <input type="text" id="name" placeholder="Ismingizni kiriting">
    </div>
    <div class="input-group">
      <label for="surname">Familiya:</label>
      <input type="text" id="surname" placeholder="Familiyangizni kiriting">
    </div>
    <div class="input-group">
      <label for="age">Yosh:</label>
      <input type="number" id="age" placeholder="Yoshingizni kiriting">
    </div>

    <div class="button-section">
      <button id="btnSave" class="btn">Saqlash</button>
      <button id="btnLoad" class="btn">Yuklash</button>
      <button id="btnclear" class="btn">Tozalash</button>
    </div>

    <hr>

    <h3>Natijalar:</h3>
    <div class="data-display">
      <p id="p_name">Ism: <b>Mavjud emas</b></p>
      <p id="p_surname">Familiya: <b>Mavjud emas</b></p>
      <p id="p_age">Yosh: <b>Mavjud emas</b></p>
    </div>

    <div id="p2">
      <!-- Dynamic content from JS -->
    </div>

    <hr>
```

```

    <h3>Dizayn Sozlamalari</h3>
    <div class="style-controls">
      <div class="control-item">
        <label for="textColor">Matn Rangi:</label>
        <input type="color" id="textColor">
      </div>
      <div class="control-item">
        <label for="color">Fon Rangi:</label>
        <input type="color" id="color" value="#f0f0f0">
      </div>
      <div class="control-item">
        <label for="fontsize">Shrift O'lchami (px):</label>
        <input type="number" id="fontsize" value="16">
      </div>
    </div>

    <div class="button-section">
      <button id="btnChange" class="btn">Dizayni Yangilash</button>
    </div>
  </div>

  <script src="script.12lab.js"></script>
</body>

</html>

```

Style.css kod:

```

/* style.css - To'liq yangilangan, tartibli va responsive dizayn uchun */

/* Boshlang'ich Sozlamalar */
:root {
  --primary-color: #2c3e50; /* Asosiy matn/sarlavha rangi (Dark Blue) */
  --accent-color: #3498db; /* Aktsent rangi (Blue) */
  --success-color: #27ae60; /* Saqlash tugmasi rangi (Green) */
  --danger-color: #e74c3c; /* Tozalash tugmasi rangi (Red) */
  --bg-light: #ecf0f1; /* Eng yengil fon (o'chirildi) */
  --bg-default: #f0f0f0; /* Umumiy fon rangi (Body uchun default) */
  --card-bg: #ffffff; /* Asosiy kontent foni */
  --border-color: #bdc3c7;
  --spacing-unit: 15px;
}

body {
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
  background-color: var(--bg-default);
  transition: background-color 0.5s ease;
  padding: 20px;
  margin: 0;
  line-height: 1.6;
  color: var(--primary-color); /* Umumiy matn rangi */
}

```

```

}

/* Asosiy Kontent Boki */
.container {
  max-width: 750px; /* Kenglikni oshirdik */
  margin: 40px auto;
  padding: 30px;
  border-radius: 12px;
  box-shadow: 0 10px 20px rgba(0, 0, 0, 0.08); /* Yumshoqroq soya */
  background-color: var(--card-bg);
}

/* Sarlavhalar */
h2, h3 {
  color: var(--primary-color);
  border-bottom: 2px solid #ecf0f1;
  padding-bottom: 10px;
  margin-top: 30px;
  font-weight: 600;
}

/* Divider */
hr {
  border: 0;
  height: 1px;
  background: var(--border-color);
  margin: 30px 0;
}

/* Input Gurutlari va Label'lar */
.input-group, .control-item {
  margin-bottom: var(--spacing-unit);
}

label {
  display: block;
  margin-bottom: 6px;
  font-weight: 500;
  color: var(--primary-color);
}

input[type="text"],
input[type="number"] {
  width: 100%;
  padding: 10px 12px;
  border: 1px solid var(--border-color);
  border-radius: 6px;
  box-sizing: border-box;
  transition: border-color 0.3s;
}

input[type="text"]:focus,

```

```
input[type="number"]:focus {
  border-color: var(--accent-color);
  outline: none;
}

/* Rang Inputlari (Kichik O'zgartirish) */
input[type="color"] {
  width: 100%; /* Konteyner ichida to'liq kenglik */
  height: 40px;
  padding: 0;
  border: 1px solid var(--border-color);
  border-radius: 6px;
  cursor: pointer;
  vertical-align: middle;
  box-sizing: border-box;
  background: none; /* Ichki fonni o'chirish */
}

/* Stilni Boshqarish Qismi (Uchta Element yonma-yon) */
.style-controls {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(180px, 1fr)); /* 3 tadan ko'proq element
bo'lsa avtomatik moslashadi */
  gap: 20px;
  margin-bottom: 20px;
}

/* Tugmalar Seksiyasi (Yonma-yon joylashuv) */
.button-section {
  margin-top: 20px;
  display: flex;
  gap: 10px;
  /* Tugmalar o'ramga sig'masa, pastga tushishini ta'minlash */
  flex-wrap: wrap;
}

.btn {
  padding: 12px 20px;
  border: none;
  border-radius: 6px;
  cursor: pointer;
  font-weight: 600;
  transition: background-color 0.3s, transform 0.1s;
  flex-grow: 1; /* Tugmalarni teng taqsimlash */
}

.btn:active {
  transform: translateY(1px); /* Bosilganda effekt */
}

/* Maxsus Tugma Stillari */
#btnSave { background-color: var(--success-color); color: white; }
```

```

#btnSave:hover { background-color: #2ecc71; }

#btnLoad { background-color: var(--accent-color); color: white; }
#btnLoad:hover { background-color: #2980b9; }

#btnChange { background-color: #f39c12; color: white; } /* "Dizayni yangilash" */
#btnChange:hover { background-color: #e67e22; }

#btnclear { background-color: var(--danger-color); color: white; }
#btnclear:hover { background-color: #c0392b; }

/* Ma'lumotlar Displayi */
.data-display {
    padding: 20px;
    border: 1px solid #dfe6e9;
    border-radius: 8px;
    background-color: #f8f8f8;
    margin-bottom: 20px;
}

.data-display p {
    margin: 8px 0;
    line-height: 1.4;
    font-size: 16px;
}

/* P2 elementi (Shrift o'zgarishi kerak bo'lgan matn) */
#p2 {
    font-weight: 700;
    padding: 10px 0;
    margin-bottom: 15px;
    font-family: Arial, sans-serif; /* JS tomonidan o'rnatiladi, lekin default qoldiramiz */
    transition: font-size 0.3s, color 0.3s;
}

/* Kichik ekranlar uchun moslashuvchanlik */
@media (max-width: 600px) {
    .container {
        padding: 20px;
        margin: 20px auto;
    }
    .button-section {
        flex-direction: column; /* Kichik ekranda tugmalar ustma-ust tushadi */
    }
    .btn {
        width: 100%;
        margin-right: 0 !important;
    }
    .style-controls {
        grid-template-columns: 1fr; /* Stil elementlari ustma-ust tushadi */
    }
}

```

```
}
```

Script.Js kod:

```
// script.js

// === DOM elementlarini tanlab olish ===
const bgElement = document.getElementById('bg');
const p2Element = document.getElementById("p2");

const nameInput = document.getElementById('name');
const surnameInput = document.getElementById('surname');
const ageInput = document.getElementById('age');

const pName = document.getElementById('p_name');
const pSurname = document.getElementById('p_surname');
const pAge = document.getElementById('p_age');

// * YANGI ELEMENT TANLANDI
const textColorInput = document.getElementById("textColor");
const colorInput = document.getElementById("color");
const fontSizeInput = document.getElementById("fontSize");

const btnSave = document.getElementById('btnSave');
const btnLoad = document.getElementById('btnLoad');
const btnChange = document.getElementById('btnChange');
const btnClear = document.getElementById('btnclear');

// === Funksiyalar ===

/**
 * Ma'lumotlarni Local Storage'ga saqlaydi
 */
function saveUserInfo() {
    const userInfo = {
        name: nameInput.value,
        surname: surnameInput.value,
        age: ageInput.value
    };

    localStorage.setItem("userInfo", JSON.stringify(userInfo));
    alert("Ma'lumotlar Local Storage'ga saqlandi!");
    loadUserInfo();
}

/**
 * Local Storage'dan ma'lumotlarni yuklaydi va displeyga chiqaradi
 */
function loadUserInfo() {
    const userInfoString = localStorage.getItem("userInfo");
```

```

let ism, familiya, yosh;

if (userInfoString) {
    const userInfo = JSON.parse(userInfoString);

    nameInput.value = userInfo.name || '';
    surnameInput.value = userInfo.surname || '';
    ageInput.value = userInfo.age || '';

    ism = userInfo.name || 'Mavjud emas';
    familiya = userInfo.surname || 'Mavjud emas';
    yosh = userInfo.age || 'Mavjud emas';

    pName.innerHTML = `Ism: <b>${ism}</b>`;
    pSurname.innerHTML = `Familiya: <b>${familiya}</b>`;
    pAge.innerHTML = `Yosh: <b>${yosh}</b>`;

} else {
    ism = 'Mavjud emas';
    familiya = 'Mavjud emas';
    yosh = 'Mavjud emas';

    pName.innerHTML = 'Ism: <b>Mavjud emas</b>';
    pSurname.innerHTML = 'Familiya: <b>Mavjud emas</b>';
    pAge.innerHTML = 'Yosh: <b>Mavjud emas</b>';
}

// p2 elementiga ma'lumotlarni chiqarish
p2Element.innerHTML = `
    **Assalomu alaykum, ${ism}!** Ma'lumotlaringiz:
    Familiya: **${familiya}**, Yosh: **${yosh}**
`;
}

/**
 * Fon rangini, tekst rangini va shrift o'lchamini o'zgartiradi va Local Storage'da saqlaydi.
 */
function applyStyleChanges() {
    // * Tekst rangi qiymatini olish va saqlash
    const newTextColor = textColorInput.value;
    localStorage.setItem("textColor", newTextColor);

    const newColor = colorInput.value;
    localStorage.setItem("bgColor", newColor);

    const newFontSize = fontSizeInput.value;
    localStorage.setItem("fontSize", newFontSize);

    // O'zgartirishlarni darhol qo'llash
    p2Element.style.color = newTextColor; // Tekst rangini qo'llash
    bgElement.style.backgroundColor = newColor;

```

```

    p2Element.style.fontSize = newFontSize + "px";

    alert(`Dizayn sozlamalari yangilandi!`);
}

/**
 * Local Storage'dagi rang va shrift sozlamalarini yuklash.
 */
function loadStyleSettings() {
    // * Tekst rangi qiymatini yuklash
    const storedTextColor = localStorage.getItem("textColor");
    const storedColor = localStorage.getItem("bgColor");
    const storedFontSize = localStorage.getItem("fontSize");

    // Tekst rangini tiklash
    if (storedTextColor) {
        p2Element.style.color = storedTextColor;
        textColorInput.value = storedTextColor;
    } else {
        p2Element.style.color = "blue"; // Boshlang'ich qiymat
    }

    // Fon rangini tiklash
    if (storedColor) {
        bgElement.style.backgroundColor = storedColor;
        colorInput.value = storedColor;
    }

    // Shrift o'lchamini tiklash
    if (storedFontSize) {
        p2Element.style.fontSize = storedFontSize + "px";
        fontsizeInput.value = storedFontSize;
    }
}

/**
 * Local Storage'ni tozalaydi va sahifani dastlabki holatga qaytaradi.
 */
function clearLocalStorage() {
    localStorage.clear();
    alert("Local Storage to'liq tozalandi!");

    // Stil sozlamalarini dastlabki holatga qaytarish uchun
    bgElement.style.backgroundColor = '#f0f0f0';
    colorInput.value = 'ffffff';
    // * Tekst rangini dastlabki holatga qaytarish
    p2Element.style.color = "blue";
    textColorInput.value = '#000000';
    p2Element.style.fontSize = '16px';
    fontsizeInput.value = '16';
    // Input maydonlarini bo'shatish
    nameInput.value = '';

```

```

    surnameInput.value = '';
    ageInput.value = '';

    loadUserInfo();
}
// === Dasturni ishga tushirish ===
loadStyleSettings();
loadUserInfo();
// === Hodisa (Event) tinglovchilarni ulash ===
btnSave.addEventListener('click', saveUserInfo);
btnLoad.addEventListener('click', loadUserInfo);
btnChange.addEventListener('click', applyStyleChanges);
btnClear.addEventListener('click', clearLocalStorage);
// Boshlang'ich stillar (fontFamily)
p2Element.style.fontFamily = "Arial";

```

Natija:

The screenshot shows a web browser window with the title "12-Laboratoriya". The address bar shows the URL "D:/3-Kurs%205-semestr/Web%20Texnologiyalar/12-laboratoriya%20amaliy/index.html". The page content is as follows:

Foydalanuvchi Ma'lumotlari

Isim: Jamshid
 Familiya: Matchanov
 Yosh: 21
 Saqlash Yuklash Tozalash

Natijalar:

Isim: Jamshid
 Familiya: Matchanov
 Yosh: 21

****Assalomu alaykum, Jamshid!** Ma'lumotlaringiz: Familiya: **Matchanov**, Yosh: **21****

Dizayn Sozlamalari

Matn Rangi:
 Fon Rangi:
 Shrift O'lchami (px): 32
 Dizayn Yangilash

Xulosa:

1. HTML (index.html) Xulosasi.

HTML fayli foydalanuvchi ma'lumotlarini kiritish va dizayn sozlamalarini boshqarish uchun a'lo darajadagi strukturani yaratgan:

- DOM uchun Maqsadli Elementlar: Har bir kiritish maydoni (name, surname, age), natija ko'rsatish joyi (p_name, p_surname), dinamik o'zgaruvchi matn joyi (p2) va barcha boshqaruv elementlari (btnSave, textColor, fontsize va h
 - Ma'lumotlar Displeyi: Kiritilgan ma'lumotlar foydalanuvchiga data-display blokida aniq ko'rsatilgan.
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- Dinamik Dizayn Boshqaruvi: Fon rangi, matn rangi va shrift o'lchamini o'zgartirish uchun `input type="color"` va `input type="number"` kabi mos elementlar ishlatilgan.

2. CSS (style.12lab.css) Xulosasi

CSS zamonaviy va tartibli dizaynni ta'minlaydi, bu esa loyihaning professional ko'rinishiga xizmat qiladi:

- Custom Properties (:root): CSS o'zgaruvchilaridan (`--primary-color`, `--accent-color`) foydalanilgan. Bu dizayn ranglarini markazlash va ularni oson o'zgartirish uchun eng yaxshi amaliyotdir.
- Flexbox va Grid: Tugmalar uchun flex va dizayn sozlamalari (`.style-c`
- UX (Foydalanuvchi Tajribasi) Detallari: Tugmalarga transition va active (bosish effekti) uslublari berilgan. Har bir tugma o'z vazifasiga mos rangga ega (Saqlash — yashil, Tozalash — qizil).

3. JavaScript (script.12lab.js) Xulosasi

JavaScript kodi laboratoriya ishi mavzusini to'liq qamrab olgan va DOM manipulyatsiyasi hamda Local Storage mexanizmlarining o'zaro ishlashini namoyish etadi.

Yakuniy Xulosa:

Ushbu laboratoriya ishi sizdan talab qilingan asosiy mavzularni (DOM orqali elementlarni o'zgartirish) mukammal qamrab olgan va Local Storage integratsiyasi orqali amaliy ahamiyatga ega bo'lgan. Siz JavaScript'ni quyidagilar uchun muvaffaqiyatli qo'lladingiz:

1. DOM O'qish: Input maydonlaridan qiymatlarni o'qish.
2. DOM Yozish/O'zgartirish: Natija matnlarini yangilash (`innerHTML`) va elementlarning rang, shrift kabi uslublarni o'zgartirish (`.style.property`).
3. Dasturni Boshlash: Sahifa yuklanganda ma'lumotlarni va dizaynni yuklash (`loadStyleSettings(); loadUserInfo();`).

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